Evidence-Led Case Review

A tax risk intelligence and investigation platform for Revenue Services Lesotho
Proposal for the RSL Taxathon, Challenge 2: Detect abusive tax planning and refund risks

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Focus	Wholesale and retail FMCG: VAT, related-party transactions, group financing and refunds
Working prototype	[prototype link], fictional data only. Sign in from the front page as any RSL officer.
RSL business owner	Deputy Commissioner, Special Sectors (proposed)

1. Executive summary

Relevant records about multinational groups are distributed across RSL systems and other sources, including income tax returns, VAT returns, the withholding tax schedule, customs, PAYE, e-invoicing, refund letters and the company registry. An officer who wants to understand one company has to assemble that picture by hand, and a recurring VAT credit or a related-party charge looks very different once the other records are next to it.

Evidence-Led Case Review joins those records per company and turns each pattern into an evidence-linked case: what was observed, which records show it, which lawful explanations may apply, what the officer must establish, and a range for the exposure. It does not decide anything. **Officers conclude; the system never does.** A pattern is a reason to look, never proof, and a refund position alone never establishes abuse.

The same system serves the three users named in the challenge: **risk analysts** (where should I look?), **refund auditors** (can I process this refund confidently?) and **Special Sectors auditors** (what is happening in this company?). It keeps legitimate refunds moving through a "lower risk" lane and keeps the evidence, the decision and the reasoning together for the next review.

This proposal describes the **full system we would deliver over a 12-month engagement**, and separately shows what a **working prototype already demonstrates**. Every capability is tagged BUILT in the prototype, PILOT (to be delivered for the officer pilot), YEAR 1 or LATER, so RSL can see exactly what is proven today and what is committed. The prototype shows that we can deliver what we propose; it is not the limit of the proposal.

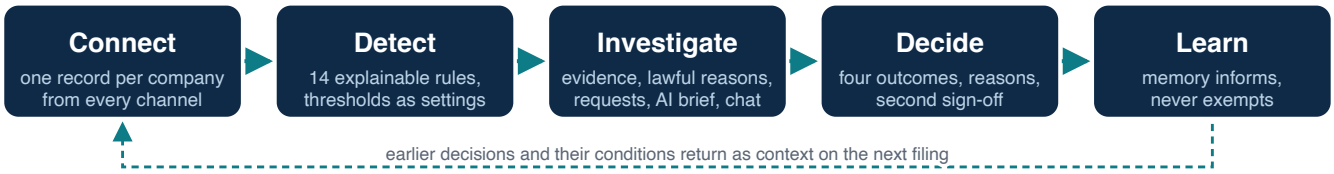


Figure 1. The five stages. Each stage keeps its evidence, so a conclusion can always be traced back to the source rows.

2. The challenge, and what we do about each part

RSL describes base erosion through related-party services and financing, in-house loss creation under the perpetual loss carry-over provision, VAT credits linked to unpaid VAT on imported services, expatriate employment income, and the line between independent consultancy and employment. The brief estimates about M1 billion in tax forgone each year; it calls this a preliminary workshop estimate, not a verified amount, and we make no claim about how much of it the system could recover.

Part of the challenge	How the system responds
Related-party fees, interest and losses that obscure profit	Rules P01–P05, P10–P12 compare fees, financing and profit with revenue and with the group register, and show the figures with the cited rows.
VAT credits from unpaid VAT on imported services	P06 compares non-resident payments in the WHT schedule with VAT declared; P07 reconciles recurring credits with invoices, imports and trading activity.
Consultancy versus employment; expatriate income	P08 flags long-term personnel arrangements and P09 unreported benefits; the officer unpacks the engagement using requested agreements and timesheets.

Legitimate refunds must not be delayed	Recurring credits that pass the checks go to a **lower risk** lane (never labelled "valid") so refund auditors can proceed.
Lawful arrangements must not be accused	Every finding lists the lawful explanations first; forbidden words (evasion, abuse, fraud) are blocked in code; wording follows the evidence available.
Legal limits on what RSL can enforce	The rules lead with enforceable checks: they cite s.113 and frame questions as allocating actual income and deductions. Pricing questions stay with the officer (AfriSam (Lesotho) (Pty) Ltd v Lesotho Revenue Authority, High Court, Commercial Division, 6 Aug 2026).

3. The system we propose

3.1 Detection: 14 explainable rules, owned by RSL

Code	Pattern	Code	Pattern
P01	Persistent losses with group charges	P08	Long-term personnel arrangement
P02	Who funds the losses	P09	Unreported fringe benefits
P03	Fees growing faster than revenue	P10	Payments to low-tax jurisdictions
P04	Related-party payments without matching WHT	P11	Large interest on group loans
P05	Treaty rate claimed	P12	Profit squeeze
P06	Imported services without matching VAT	P13	Information gap (not a risk finding)
P07	Recurring VAT credits (lower-risk lane)	P14	Possible data capture error

Each rule states what it examines, which records it needs, its threshold, why it matters, the lawful explanations, and the documents an officer may request. Thresholds are versioned settings, not code: RSL's rule owner changes them with a recorded reason, and a rule version is never edited after use PILOT: rule-owner editing screen. Missing records produce an **information gap** (P13), never a suspicion. A figure that looks like a keying error goes to **data review** (P14), never into company memory.

3.2 Evidence controls, enforced in code BUILT

- **Evidence ladder.** A return alone: "may need examination". Return plus other RSL records: "appear inconsistent with each other". Plus outside records: "the group structure suggests". Plus taxpayer documents: "does not appear to match the evidence provided". The conclusion belongs only to the officer.
- **Every finding shows** the observation, the lawful explanations, what the officer must establish, an estimated exposure *range* (never an assessed amount) and links to the exact source rows.
- **Sequencing.** A taxpayer document stays invisible until an officer sends an information request naming it; it arrives after the stated response time.
- **Conclusions are stored as five parts:** observed pattern, conclusion, evidence, conditions, reassessment point.

3.3 The officer's workspace

Each user group has its own desk and worklist (BUILT): analysts receive new filings, refund auditors see VAT credits and the lower-risk lane, Special Sectors auditors receive referred cases. A case carries its findings, evidence, a checklist, requests, documents, notes and a timeline; cases are ranked by estimated exposure and routed by rule. The officer records one of four outcomes, each with a reason: **Concern supported, referred** (not a finding of wrongdoing), **Explained, legitimate**, **More information needed**, or **System error**. Closing a finding as legitimate at or above **M1,000,000** estimated exposure (configurable) needs a second officer's sign-off. Planned additions: **officers adding a finding the rules missed** and editing brief notes in place PILOT.

3.4 Company memory: informs, never exempts BUILT

Earlier decisions and their conditions are kept with the company. A later finding is still raised, with the earlier decision shown as context, and the system states when a pattern has moved outside the conditions of a decision. In the prototype, a fee increase at Qiloane is shown against decision DEC-0007, with the statement that the new fees fall outside its conditions.

3.5 AI assistance, grounded and optional BUILT

- **Case brief.** The application builds a fact sheet of records it has looked up; the model writes a brief from that sheet only, each statement citing record IDs. A guard rejects any statement that cites an unknown ID, cites nothing, or breaks the wording policy, and a plain template brief takes over so the case page never fails. The officer approves, edits or rejects it.
- **Ask the records.** An officer asks a question ("which companies had related-party fees rise more than 30%?"); the system finds the records first, the model may only word an answer from them, and every answer cites its sources. The guard also rejects any figure not in the facts, and a lookup answer is the fallback. It is a way into the same records, never a separate source.
- **Already working with a local model.** The prototype is integrated with a locally hosted model (gpt-oss 20B) running on my own Mac mini M4 through Ollama. The web application reaches it over a private tunnel, and the case briefs and Ask the records described above are generated by it today, on fictional data.
- **Data stays inside RSL.** The same local-processing design is intended for RSL's own infrastructure, so no taxpayer record leaves RSL. An approved cloud model is an option only if RSL's policy allows it; the provider is a setting.
- Stronger numeric cross-checking of brief statements, and automatic redrafting when findings change PILOT. Semantic search across taxpayer documents YEAR 1.

3.6 Audit and notifications BUILT

Every receipt, analysis, case opening, request and conclusion is written to a **tamper-evident audit log**: entries are hash-chained, the database refuses edits and deletions, and a command verifies the chain. No blockchain is needed. Hand-offs send a **WhatsApp** message that carries no taxpayer details, only a case reference and priority, pointing the officer into the system.

3.7 Beyond the pilot

YEAR 1 A secure taxpayer response portal; scheduled feeds from eTax, eCustoms and Lekuka in place of file uploads, once RSL agrees; management reports on findings, outcomes and refund turnaround. LATER A data warehouse for history across tax types and years, and forecasting or exposure modelling once enough reliable outcomes exist. Potential exposure, assessed liability and collected revenue stay separate measures; none of this is claimed for the prototype.

4. Three cases through the system

The brief asks for an ordinary case, an exception and a case needing human review. The prototype walks each one end to end.

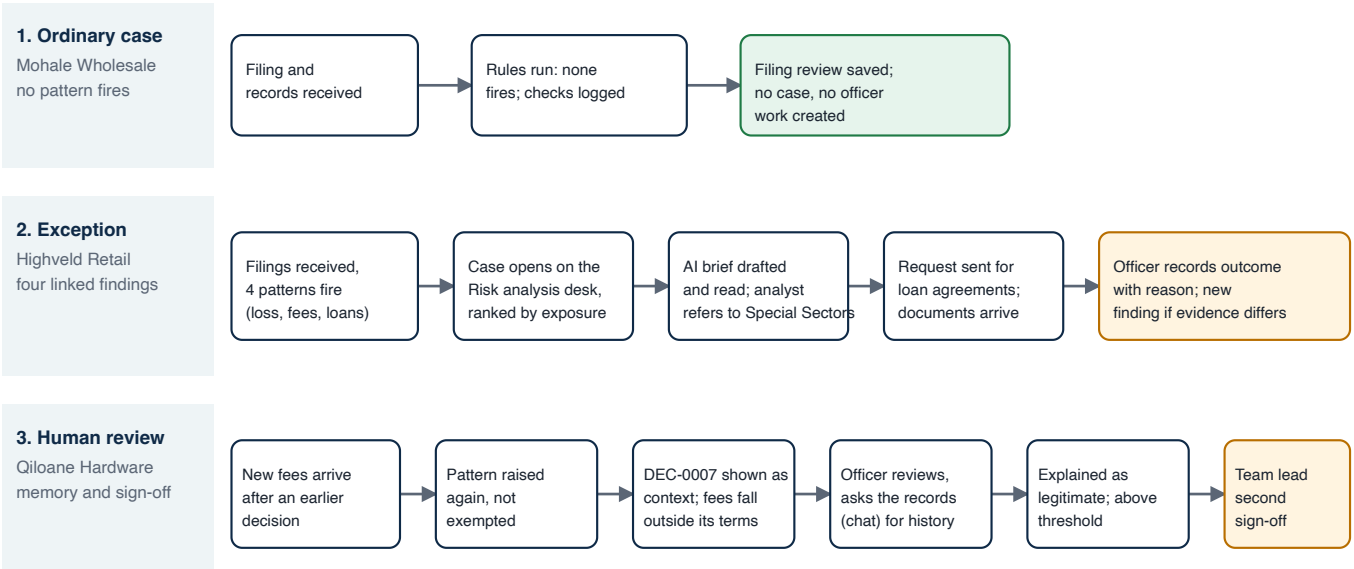


Figure 2. Three journeys. Green: ordinary result. Amber: a human conclusion is recorded. In every case the system proposes and an officer decides. A fourth path, a figure that looks wrongly captured (Lebelo), goes to data review and never into company memory.

5. Architecture, data and design for Lesotho

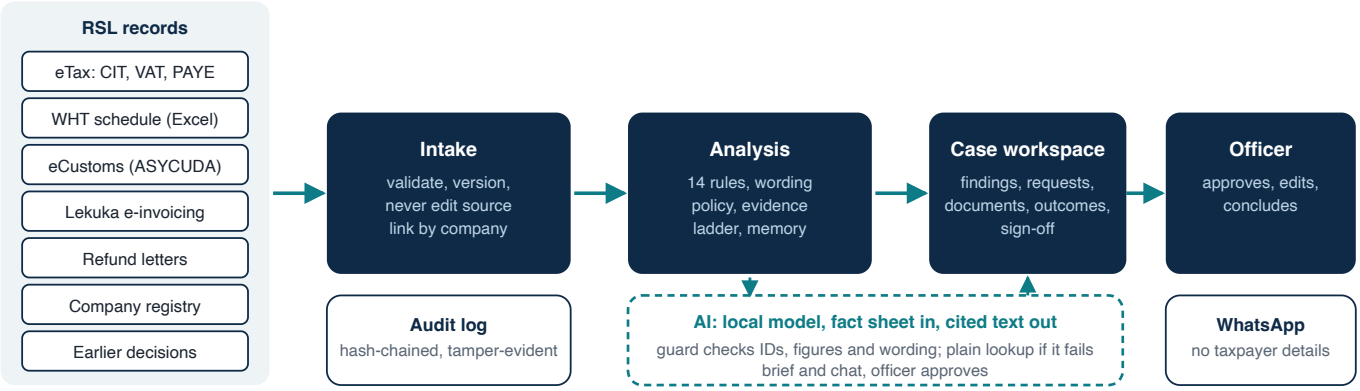


Figure 3. Records flow in once, stay unedited, and every claim downstream cites them. The model sees only the compact fact sheet for one case, never the database.

Data required

Confirmed with RSL in discovery; pilot uses approved extracts or masked history, not live access.

- CIT and VAT returns; PAYE; WHT payment schedules
- Customs imports; Lekuka monthly summaries
- Refund claim letters; registration and group register
- Past officer decisions, to seed company memory

Dependencies: RSL IT (eTax, eCustoms, Lekuka owners), the companies registry for group data, Meta for WhatsApp approval, and RSL hosting policy.

Designed for Lesotho

- **Limited connectivity:** light server-rendered pages, no heavy front end, resumable uploads PILOT
- **Language:** officer screens in English; Sesotho versions of notifications and taxpayer information-request letters PILOT
- **Record keeping:** intake reads the formats RSL already uses (CSV, the emailed WHT Excel schedule, scanned letters)
- **Explainable:** every statement cites a record; every outcome carries its reason

6. Pilot proposal

Element	Proposal
Business owner	Deputy Commissioner, Special Sectors, with a named working lead from each of the three teams.
User group	About 12 officers: 4 risk analysts, 4 refund auditors, 4 Special Sectors auditors, plus 1–2 team leads for second sign-off.
Taxpayers	An agreed list of wholesale and retail FMCG taxpayers with related-party transactions or recurring VAT credits (about 40–60), using approved extracts.
Test period	Back-test on masked history in months 5–6; live officer pilot of 12 weeks in months 7–9 (Jun–Aug 2027), after a baseline taken in months 2–3.
Baseline	Measured from RSL's own records: time to review a case, time to assemble evidence, refund processing time, share of cases escalated and outcomes of earlier reviews.
Success measures	Findings supported by valid cited records; false positives and missed patterns on the back-test; review time and evidence-assembly time against baseline; **no legitimate refund delayed**; share of cases needing extra information; officer adoption and usefulness of the brief and chat; completeness of recorded decisions. Targets are agreed with RSL before the pilot starts.
Test immediately	Officer walk-through of the prototype; rule workshops on the 14 patterns; back-test on masked historical records.
Needs agreement or change	Written data-sharing authority and extract formats; registry data-sharing agreement; hosting and AI-location decision; WhatsApp approval for operational use; confirmation of the legal basis cited on each rule; ownership and licence terms for the platform and RSL's rules and data.

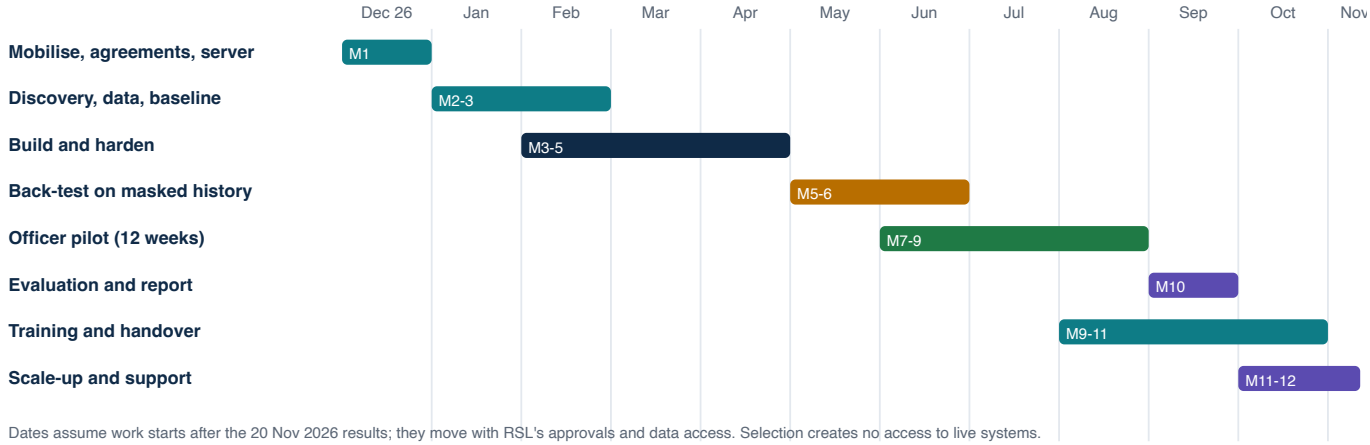


Figure 4. The 12-month plan. The pilot starts only after the back-test, the security test and the data and hosting agreements are in place.

7. Resources, budget and operating cost

I am submitting as an individual and will lead and architect the work. Skills are brought in **when a milestone needs them**, part-time, not carried for the whole year. All figures are in Maloti and are estimates for discovery to confirm; they are not a request for payment, and selection creates no procurement commitment.

7.1 Who is needed, and when

Role	When	Effort	What they deliver
Senior developer and architect (lead)	Months 1–12	full time	Architecture, rules, AI integration, delivery, RSL liaison
Junior developers (2)	Months 1–12	full time	Pipelines, screens, tests, documentation, support
Linux server engineer	Month 1, then monthly	8 days + retainer	Build, harden and back up the Linux server; monitoring; updates
UX and product designer	Months 2–5	20 days	Officer workflows, usability testing with RSL staff, Sesotho content
International tax and VAT specialist	Months 2–6	24 days	Rule workshops, thresholds, back-test review, wording review
Data-protection and legal adviser	Months 1 and 6	6 days	Data-sharing terms, retention, pilot safeguards
Independent security tester	Month 6	fixed	Penetration and configuration test before the pilot
Trainer and technical writer	Months 9–11	8 days	Officer training, administrator handover pack

RSL contributes: a business owner and working leads, data extracts, about 12 officers for the pilot, access to hosting, and prompt decisions on the agreements above.

7.2 Year-one budget

Item	Basis	Amount (M)
Senior developer and architect	M26,000 × 12 months	312,000
Two junior developers	M10,000 × 2 × 12 months	240,000
Claude coding subscription	3 seats × 12 months × about M1,900	68,400
UX and product designer	20 days × M1,500	30,000
Linux server engineer	8 days setup, then M1,500 a month	30,000
International tax and VAT specialist	24 days × M2,500	60,000
Data-protection and legal adviser	6 days × M3,000	18,000
Independent security test	Fixed	20,000

Trainer and technical writer	8 days × M1,500	12,000
Linux server, backups, domain and SSL	12 months	25,500
Local AI inference machine	One-off allowance	45,000
WhatsApp messaging	Allowance	3,000
Subtotal		863,900
Contingency	10%	86,390
Year-one total		**950,290**



The coding subscription is a development tool. It is used only with fictional or masked data, never with taxpayer records. The subscription price and exchange rate are assumptions to confirm.

7.3 Operating cost after year one (estimate)

Item	Per year (M)
One junior developer for maintenance and support, plus about 20% of the lead's time	about 182,400
Server, backups and server-engineer retainer	about 42,000
WhatsApp messaging and licences	about 6,000
Annual run cost	**about M230,000**

A locally hosted model has no per-use fee. If RSL prefers an approved cloud model, a usage cost replaces part of the hardware allowance.

8. Risks, honesty about results, and ownership

Risk	Response
Data arrives late or inconsistent	Discovery and profiling in months 2–3; information-gap findings instead of guesses.
False positives burden officers	Explainable rules, lawful explanations shown first, back-test against baseline, rule owner adjusts thresholds.
AI states something untrue	Grounded fact sheet, citation, figure and wording guard, template fallback, officer approval. The case-brief guard checks citations, not the truth of every sentence; stronger checks are a pilot deliverable.
Disclosure of taxpayer data	Local model by default, no taxpayer details in notifications, role-based access, audit log, security test before the pilot.
One person leads the work	Two juniors work alongside throughout; documentation and a handover pack; code and configuration available to RSL.

> **Honest about results.** The prototype's data was written to contain known stories. It shows the workflow and the rules behave as designed: the 17 expected findings are produced with the expected records, the two ordinary companies and three lawful decoys raise nothing (apart from one lower-risk VAT result), and a full automated test suite passes. In a trial on 10 open cases with a local model, 9 briefs passed the guard first time and the guard caught the tenth. **This does not show detection accuracy on real taxpayers.** Only the pilot can.

Ownership. The call states that ideas generated at the event remain RSL's intellectual property. We accept that RSL owns its rules, thresholds, data, decisions and outputs; the licence terms for the platform software would be agreed with RSL before the pilot.

9. Value and how this meets the criteria

Criterion	How the proposal responds
Taxpayer and business value, 25%	Officers see one evidence-linked picture per company; legitimate refunds move through a lower-risk lane; nothing is accused.
Performance and evidence, 20%	Rule results match the expected findings with cited rows; baseline, back-test and measured pilot; limits stated.
Usability, 15%	Desks by role, one case workspace, a guided tour, plain language, a records chat.
Trust and compliance, 15%	Wording and evidence ladder in code, human decisions, second sign-off, tamper-evident log, no details in notifications, local AI.
Pilot potential, 15%	Working prototype, a 12-month plan, a costed team and an annual run cost.
Innovation, 10%	A risk carries its evidence and reasoning from detection to conclusion, with memory that informs and never exempts.

The aim is not to replace the judgment of tax professionals. It is to give them better-connected information, clearer evidence and a consistent way to carry an investigation forward.

Annex A. Capability map: what is built, and what is proposed

Status shows what is demonstrated in the prototype today and when the remainder is delivered. Nothing marked Pilot, Year 1 or Later is claimed as built.

Capability	Status	Notes
Intake of every channel (CIT, VAT, WHT, PAYE, customs, Lekuka, refund letters, registry, decisions)	BUILT	Validated file import; corrections are new versions, never edits; one analysis run per batch.
14 detection rules; findings with cited rows, exposure range and lawful explanations	BUILT	Match the expected findings word for word on the test data.
Wording policy and evidence ladder; information-gap and data-error paths	BUILT	Applies to AI output too.
Cases, routing, role desks, worklists, ranking by exposure	BUILT	Risk analysis, Refund review, Special Sectors, team lead, data team.
Information requests and sequenced taxpayer documents	BUILT	Documents appear only after a matching request and the response delay.
Four outcomes, five-part conclusions, second sign-off	BUILT	Threshold configurable; the demo uses a lower one.
Company memory with conditions and reassessment point	BUILT	Qiloane and DEC-0007; system errors kept out of memory.
AI case brief (grounded, guarded, template fallback, officer review)	BUILT	Local model through Ollama; cloud provider optional.
Ask the records (cited answers, figure check, lookup fallback)	BUILT	Page and panel on each case. Runs on the local model; answer quality to be measured with officers.
Tamper-evident audit log with verification command	BUILT	Hash chain plus database protection.
WhatsApp notifications without taxpayer details	BUILT	Four approved templates; verified test number only.
Guided tour, officer front page, resettable demo	BUILT	11-step tour; nightly reset.
Rule-owner screen to edit thresholds with a recorded reason	PILOT	Thresholds are already versioned settings.
Officer adds a finding the rules missed; edit notes in place	PILOT	Feeds catalogue review.
Numeric cross-check of brief statements; redraft when findings change	PILOT	Closes known limits of the guard.
Registry and group matching on real extracts; scheduled feeds	PILOT	Approved extracts first.
RSL sign-in, role administration, production hosting and security test	PILOT	Linux server, backups, monitoring.
Sesotho notifications and letters; low-bandwidth refinements	PILOT	Tested with officers.
Secure taxpayer response portal	YEAR 1	Replaces emailed documents.
Direct integrations with eTax, eCustoms, Lekuka	YEAR 1	Subject to RSL agreement.
Management reports; semantic search across documents	YEAR 1	Outcomes, turnaround, rule performance.
Data warehouse; forecasting and exposure modelling	LATER	Needs reliable outcomes history; exposure is never shown as revenue.